

Lab Manual

Programming Fundamental



University of Haripur

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Note: At the end of this lab manual there is a Lab evaluation summary sheet. It is the responsibility of every student to get that sheet signed from his/her lab engineer after every lab. If He / She fail to do so, that particular unsigned lab will be marked zero.

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INTRODUCTION

The objective of this lab manual is to give students step-by-step examples to become familiar with programming concepts, design, and coding.

FEATURES

To ensure a successful experience for instructors and students alike, these lab manuals includes the following features:

- **Lab Objectives**—Every lab has a brief description and list of learning objectives
- **Materials Required**—Every lab includes information on hardware, software, and other materials you will need to complete the lab
- **Completion Times**—Every lab has an estimated completion time so that you can plan your activities more accurately
- **Activity Sections**—Labs are presented in manageable sections; where appropriate, additional Activity Background information is provided to illustrate the importance of a particular project
- **Step-by-Step Instructions**—Every lab provides steps to enhance technical proficiency; some labs include Critical Thinking exercises to challenge students
- **Review Questions**—Some labs include review questions to help reinforce concepts presented in the lab

SOFTWARE REQUIREMENTS

- Computer running Windows 10, Windows 7 etc.
Recommended compiler is Dev C++

COMPLETING THE LAB ASSIGNMENTS

Some lab assignments require written answers to complete an exercise, while others are programming assignments that require you to work with a C++ compiler.

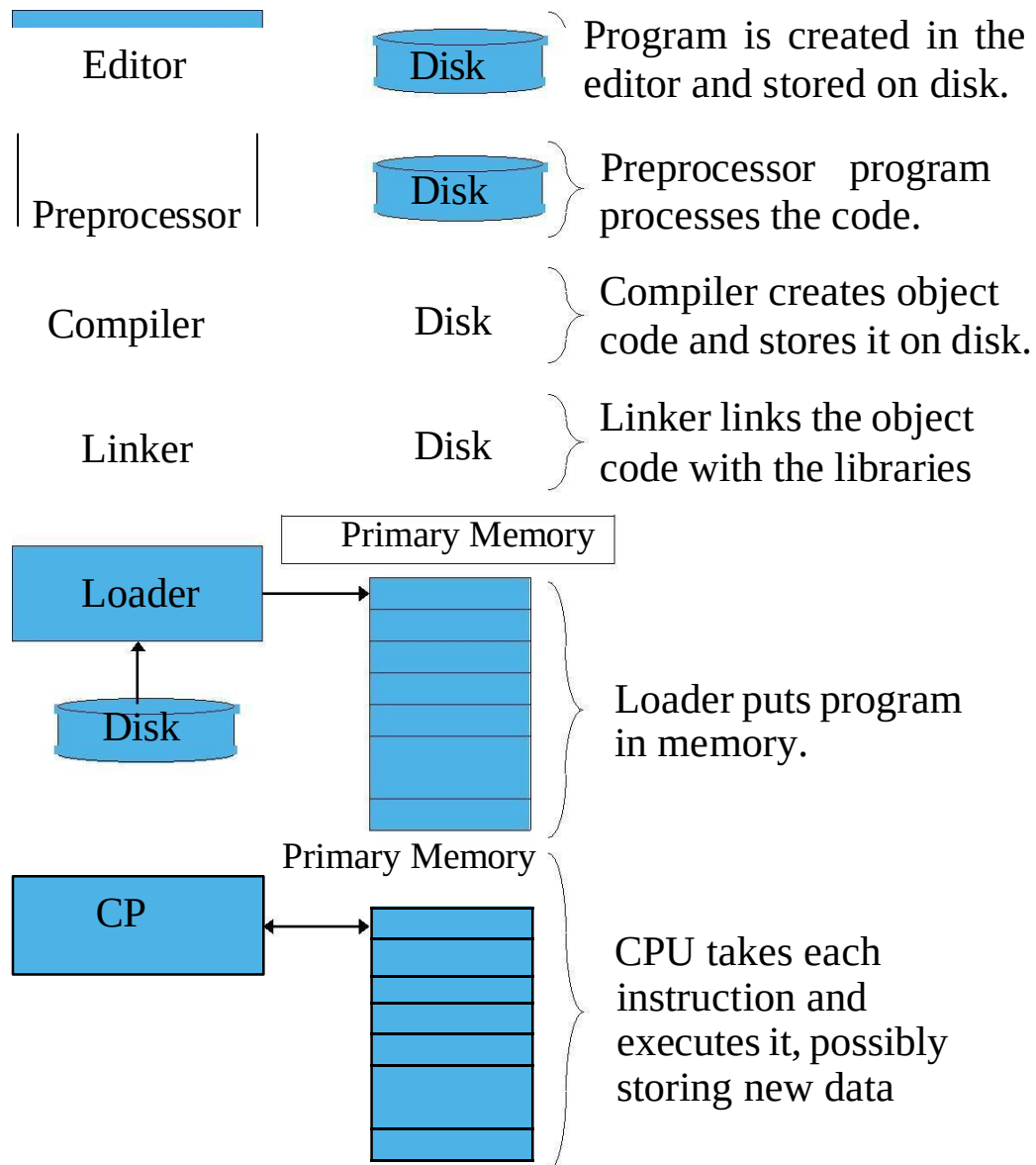
- Check with your instructor for instructions on completing the written assignments. For example, you can print pages directly from the appropriate PDF file, and then write directly on the page.
- To complete the programming assignments, use the compiler that your instructor recommends or requires. Print all the documentation assigned, including program code, program prompts, input, and output displayed on the screen, input files, and output files. You can submit your written answers and the printed documentation with a lab cover sheet for grading. If your instructor requires an electronic copy of your work, e-mail the completed assignment to your instructor or include a removable disk with your work. Your instructor will tell you what is

needed, but be sure to submit the .cpp, .h, and any .srt or .txt files that you create, as well as any input and output files. Also include your name or ID in the titles of all your files. To provide program documentation, compile and run your program, copy the prompts, input, and output (if appropriate), and paste them as a block comment at the end of your program. Use the Copy and Paste features of your C++ program development kit to do so. After you paste the comment in the program, either print the program file from your text editor or submit the program file to your instructor electronically.

Lab Manual # 1

Basic

1.1 A typical C++ environment



1.2 Program No 1

```
#include<iostream>
main ( )
{
    cout << — Welcome to —;
}
```

1.2.1 Out Put of Program No 1

Output
Clear

```
ERROR!
/tmp/NSdbFxu1sC.cpp:4:9: error: extended character — is not valid in
an identifier
  4 | cout << — Welcome to —;
    |           ^
/tmp/NSdbFxu1sC.cpp:4:23: error: extended character — is not valid in
an identifier
  4 | cout << — Welcome to —;
    |                   ^
/tmp/NSdbFxu1sC.cpp:2:1: warning: ISO C++ forbids declaration of
'main' with no type [-Wreturn-type]
  2 | main ( )
    | ^~~~
ERROR!
/tmp/NSdbFxu1sC.cpp: In function 'int main()':
/tmp/NSdbFxu1sC.cpp:4:1: error: 'cout' was not declared in this scope
; did you mean 'std::cout'?
  4 | cout << — Welcome to —;
```

1.3 Declare Variables

Declare 3 Integer Type & 3 float type Variables.

<i>INT</i>	<i>Float</i>
10	1.5
20	2.75
30	3.14159

1.4 Program No 2 Example Of Assignment Operator

```
#include <iostream>
using namespace std;
int main ()
```

```
{  
  int a, b;  
  a = 10;  
  b = 4; a  
  = b;  
  b = 7;  
  cout << "a :";  
  cout << a;  
  cout << " b :";  
  cout << b;  
  return 0;  
}
```

1.4.1 Out Put of Program No 2

```

Output
/tmp/sYcU8VnFJk.o
a :4 b :7

=== Code Execution Successful ===

```

1.5 Arithmetic Operators

Write Arithmetic Operators

Addition	(+): Adds two numbers
Subtraction	(-): Subtracts one number from another.
Multiplication	(*): Multiplies two numbers.
Division	(/): Divides one number by another. Integer division truncates the result to an integer.

1.5.1 Precedence

Write precedence of the arithmetic operators

Parenthesis:	operations within the parentheses are always evaluated first
Exponentiation:	^(not used in this example)
Multiplication, division, modules:	are evaluated from left to right
Addition:	are evaluated from left to right
Subtraction:	are also evaluated from left to right

1.6 Program No 3

Write a program in C++ that display following output

```

*****
*****
*****
*****
*****
*****
*****
*****
*****
*****
*****
*****
#####
#####
#####

```

1.6.1 Code of Program No 3

```

#include <iostream>
using namespace std;
int main() {

    int a, b;
    a= 10;
    b= 4;
    a= b;
    b=7;
    cout << "a: " << a <<
    endl;
    cout << "b: " << b <<
    endl;
    return 0;}

```


1.7 Program No 4

Write a program in C++ which prints following output,

```
$*****$
$*****$
$*Welcome to          *$
$*   School of Engineering  *$
$*       Isamabad.        *$
$*****$
$*****$
```

1.7.1 Code of Program No 4

```
#include <iostream>
using namespace std;
int main() {
    for (int i = 0; i < 10; i++) {
        cout << "*****\t*****" << endl;
    }
    cout << endl;
    cout << "*****\t*****" << endl;
    for (int i = 0; i < 3; i++) {
        cout << "*****\t*****" << endl;
    }
    return 0;
}
```

1.8 Program No 5

Write a program in C++, which takes radius from the user and calculate the area of

sphere i.e
 $\text{Area} = 4\pi r^2$

(Hint $\pi = 3.1416$
 $\text{Area} = 4 * 3.1416 * r * r$)

1.8.2 Code of Program No 5

```
#include <iostream>
using namespace std;
int main() {
    const double pi = 3.1416;
    double radius, area;
    cout << "Enter the
radius of the sphere: ";
    cin >> radius;
    area = 4 * pi * radius *
radius;
    cout << "The area of
the sphere is: " << area
<< endl;
    return 0;
}
```

1.8.3 Out Put of Program No 5

1.9 Program No 6

Write a program to find the number of bytes occupied by various data types using the sizeof operator?

```
int a;  
char b;  
float c;  
long int  
d; bool e;  
unsigned int j;  
unsigned long k;
```

1.9.1 Code

```
#include <iostream>  
using namespace std;  
int main() {  
    int a;  
    char b;  
    float c;  
    long int d;  
    bool e;  
    unsigned int j;  
    unsigned long k;  
    cout << "Size of int: " << sizeof(a) << " bytes" <<  
endl;  
    cout << "Size of char: " << sizeof(b) << " bytes"  
<< endl;  
    cout << "Size of float: " << sizeof(c) << " bytes"  
<< endl;  
    cout << "Size of long int: " << sizeof(d) << "  
bytes" << endl;  
    cout << "Size of bool: " << sizeof(e) << " bytes"  
<< endl;  
    cout << "Size of unsigned int: " << sizeof(j) << "  
bytes" << endl;  
    cout << "Size of unsigned long: " << sizeof(k) << " bytes" << endl;    return 0;  
}
```


Lab Manual # 2

if / if-else Statements

2.1 Que No 1 Largest value among three

Write a program in C++ that take input of three integer's numbers from user. Find the largest number among three of them.

2.1.1 Code Area

```
#include <iostream>
using namespace std;
int main() {
    cout << "Hello, this is Program 1 of Lab Manual 2.\n";
    int num1, num2, num3;
    cout << "Enter three numbers: ";
    cin >> num1 >> num2 >> num3;
    if (num1 >= num2 && num1 >= num3) {
        cout << "The largest number is " << num1 << endl;
    }
    else if (num2 >= num1 && num2 >= num3) {
        cout << "The largest number is " << num2 << endl;
    }
    else {
        cout << "The largest number is " << num3 << endl;
    }
    return 0;
}
```


2.2 Que No 2 Grade Program using nested if else

Write a program in C++ using if/else operator with nested statements to find the grade of a student.

The detail is as follow.

grade >= 90	➔	Grade A
grade >= 80	➔	Grade B
grade >=70	➔	Grade C
grade >=60	➔	Grade D

2.2.1 Code Area

```
#include <iostream>
using namespace std;
int main() { int grade;
    cout << "Enter the grade of the
student:";
    cin >> grade;
    if (grade >= 90) {
        cout << "Grade A" << endl;
    }
    else {
        if (grade >= 80) {
            cout << "Grade B" << endl;
        }
        else {
            if (grade >= 70) {
                cout << "Grade C" << endl;
            }
            else {
                if (grade >= 60) {
                    cout << "Grade D" << endl;
                }
                else {
                    cout << "Grade F" << endl;
                }
            }
        }
    }
    }
    }return 0;
}
```


2.3 Que No 3 vowel / Consonant

3.1.1 Code

```
char ch;
string vowels = "aeiouAEIOU";
cout << "Enter a character: ";
cin >> ch;
if (vowels.find(ch) != string::npos) {
    cout << "It is a vowel" << endl;
}
else {
    cout << "It is a consonant" << endl;
}
```

2.4 Que No 4 Even / Odd

Write a Program in C++ that take an Integer Value from the user and tell that the number is EVEN or ODD.

2.4.1 Code Area

```
#include <iostream>
using namespace std;
int main() {
    int num;
    cout << "Enter an integer: ";
    cin >> num;
    if (num % 2 == 0) {
        cout << "The number " << num << " is
EVEN." << endl;
    } else {
        cout << "The number " << num << " is
ODD." << endl;
    }

    return 0;
}
```


2.5 Que No 5 Capital / Small Letter

Write a program in C++ that take a single character from the user, and tells it's a Small Letter or it's a CAPITAL letter using nested if statement only.

2.5.1 Code Area

```
#include <iostream>

using namespace std;

int main() {
    char ch;
    cout << "Enter a character: ";
    cin >> ch;
    if (ch >= 'A' && ch <= 'Z') {
        cout << "It is a CAPITAL letter." << endl;
    }
    else {
        if (ch >= 'a' && ch <= 'z') {
            cout << "It is a Small letter." << endl;
        }
        else {
            cout << "It is not a letter." << endl;
        }
    }
    return 0;
}
```


2.6 Que No 6 Ice / water / steam

Make a program in C ++ that tells the form of Water whether it is Ice, Water or Steam. Display the menu also as under.

Temperature Less than 0 = ICE

Temperature Greater than 0 & Less than 100 = Water

Temperature Greater than 100 = STEAM

2.6.1 Code Area

```
#include <iostream>

using namespace std;

int main() {
    float temperature;
    cout << "Water State Menu:\n";
    cout << "1. Temperature less than 0: ICE\n";
    cout << "2. Temperature greater than 0 and less than 100: WATER\n";
    cout << "3. Temperature greater than 100: STEAM\n";
    cout << "-----\n";
    cout << "Enter the temperature: ";
    cin >> temperature;
    if (temperature < 0) {
        cout << "The form of water is ICE.\n";
    }
    else if (temperature >= 0 && temperature < 100) {
        cout << "The form of water is WATER.\n";
    }
    else {
        cout << "The form of water is STEAM.\n";
    }
    return 0;
}
```


Lab Manual # 3

Switch Statement

3.1 Que No 1 Switch statement

Write a program in C++ using switch statement that contain option as under

Enter 1--> To Find Largest Number Among Three Variables.

Enter 2--> To Find ODD or EVEN

Enter 3--> To Find Condition of Water

Enter 4--> To Find Grade Of Student

Detail of Option 3

Temperature Less than 0 = ICE

Temperature Greater than 0 & Less than 100 = Water

Temperature Greater than 100 = STEAM

Detail of option 4

grade >= 90	➔	Grade A
grade >= 80	➔	Grade B
grade >=70	➔	Grade C
grade >=60	➔	Grade D

3.1.1 Code Area

```
#include <iostream>
```

```
using namespace std;
```

```
{  
  
    int choice;  
  
    cout << "Menu Options:\n";  
    cout << "1 --> To Find Largest Number Among Three  
Variables.\n";  
    cout << "2 --> To Find ODD or EVEN.\n";  
    cout << "3 --> To Find Condition of Water.\n";  
    cout << "4 --> To Find Grade of Student.\n";  
    cout << "Enter your choice: ";  
    cin >> choice;  
  
    switch (choice) {  
        case 1: {  
            int a, b, c;  
            cout << "Enter three numbers: ";  
            cin >> a >> b >> c;  
  
            if (a >= b && a >= c)  
                cout << "The largest number is: " << a << endl;  
            else if (b >= a && b >= c)  
                cout << "The largest number is: " << b << endl;  
            else  
                cout << "The largest number is: " << c << endl;  
            break;  
        }  
  
        case 2: {  
            int num;  
            cout << "Enter a number: ";  
            cin >> num;  
  
            if (num % 2 == 0)  
                cout << "The number " << num << " is EVEN.\n";  
            else  
                cout << "The number " << num << " is ODD.\n";  
            break;  
        }  
  
        case 3: {  
            float temp;  
            cout << "Enter the temperature: ";  
            cin >> temp;  
  
            if (temp < 0)  
                cout << "The form of water is ICE.\n";  
            else if (temp >= 0 && temp < 100)  
                cout << "The form of water is WATER.\n";  
            else  
                cout << "The form of water is STEAM.\n";  
            break;  
        }  
  
        case 4: {  
            int grade;
```

```
cout << "Enter the grade of the student: ";
cin >> grade;

if (grade >= 90)
    cout << "Grade A\n";
else if (grade >= 80)
    cout << "Grade B\n";
else if (grade >= 70)
    cout << "Grade C\n";
else if (grade >= 60)
    cout << "Grade D\n";
else
    cout << "Grade F\n";
break;
}
default:
    cout << "Invalid choice! Please enter a valid option.\n";
return 0;
}
```


Lab Manual # 4

For/while Loops

4.1 Example

```
int sum = 0;

for (int i = 1; i <= 10; i++) {

    sum += i;

}

cout << "The sum of integers from 1 to 10 is: " << sum << endl;
```

4.1.1 Output

Sum=55

4.2 Example Of While Loop

```
int sum ,
number ; sum =
0 ;
number = 1 ;
while ( number <= 1000 )
{
    sum = sum + number ;
    number = number + 1 ;
}
cout << — The sum of the first 1000 integer starting from 1 is || << sum ;
```

4.2.1 Output

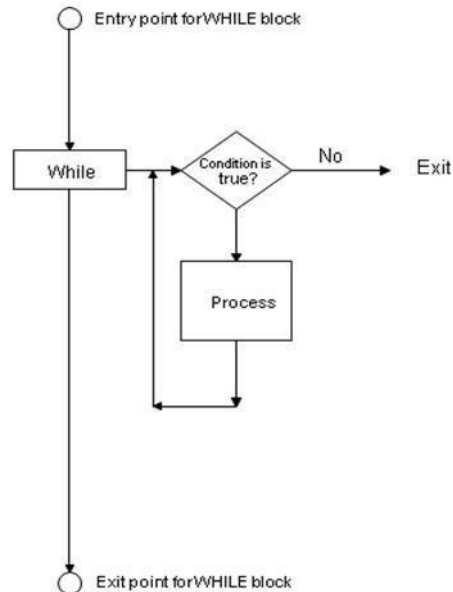
This code will output:

```
The sum of the first 1000 integers
starting from 1 is 500500
```


4.3 Flow chart of while Loop

Flow Chart for While Construct

WHILE Statement



4.4 Factorial Definition

$$n! = n*(n-1)*(n-2)*(n-3).....*3*2*1$$

4.4.1 Code of Factorial Program

```

int number ;
int factorial ;
factorial = 1 ;

cout << —Enter the number of
Factorialll ; cin >> number ;
while ( number >= 1 )
{
    factorial = factorial *
    number ; number = number –
    1 ;
}
cout << —Factorial isll << factorial ;
  
```

Out Put

4.5 For loop

```
for ( initialization condition ; termination condition ; increment condition )
{
    statement ( s ) ;
}
```

4.5.1 Example of for loop

```
int counter ;
for( counter = 0 ; counter < 10 ; counter = counter +
    1 ) cout << counter;
```

4.5.2 Write the output of the program step by step

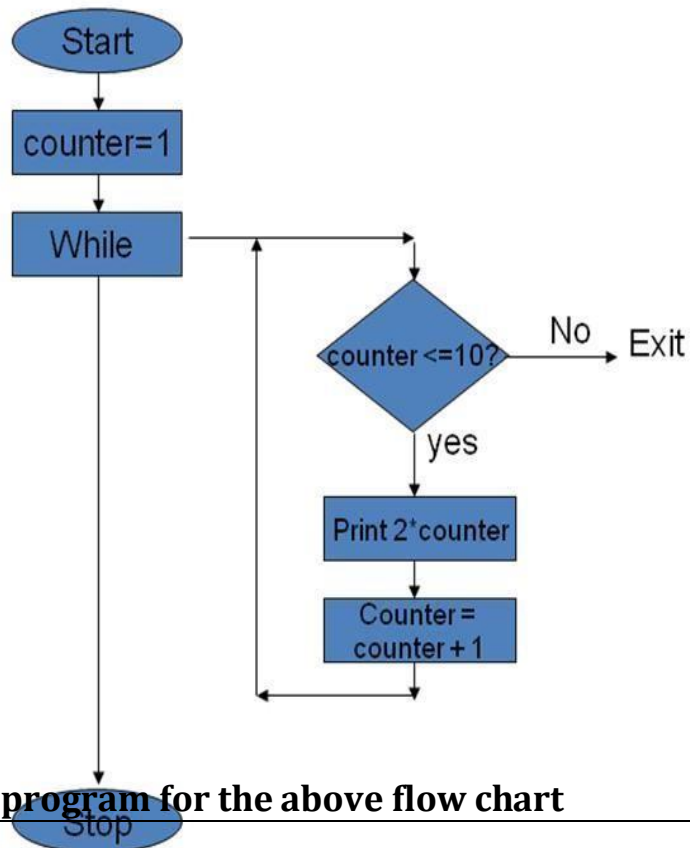
☐ Initialization (counter = 0): The loop starts with counter initialized to 0. ☐ Output (cout << counter): 	
The current value of counter is printed to the console. ☐ Increment (counter = counter + 1):	
After printing, counter is incremented by 1. ☐ Repeat:	
Steps 2–4 repeat for counter values 1 to 9. ☐ Exit:	
When counter = 10, the condition counter < 10 becomes false, and the loop exits	

4.5.3 Final Output

0123456789

4.6 Flow Chart of Table 2

Flow chart for the 'Table' example



4.6.1 Write a program for the above flow chart

```
#include <iostream>
using namespace std;
int main() {
    int counter = 1;
    while (counter <= 102) {
        cout << 2 * counter << endl;
        counter++;
    }
    return 0;
}
```


4.7 Example

Write a Program That Display numbers from
1,2,3,4,5,6,7,8,9..... 10000

4.7.1 Code

```
for (int i = 1; i <= 10000; i++) {  
    cout << i << " ";  
}
```

4.8 QUE NO 1 Terminate program on entering zero

Write a Program in C++ that Exit on entering 0, Using While Loop.

4.8.1 Code

```
#include <iostream>  
using namespace std;  
  
int main() {  
    int number;  
    cout << "Enter numbers (enter 0 to exit): " << endl;  
    while (true) {  
        cin >> number;  
        if (number == 0) {  
            cout << "Exiting the program." << endl;  
            break;  
        }  
    }  
}
```

```
    }  
    cout << "You entered: " << number << endl;  
}  
  
return 0;  
}
```


4.9 QUE NO 2 Factorial program

Write a Program in C++ that calculate the factorial of a user defined number, using for loop

4.9.1 Code

```
#include <iostream>

using namespace std;

int main() {
    int number;
    int factorial = 1;
    cout << "Enter a number to calculate its factorial: ";
    cin >> number;
    if (number < 0) {
        cout << "Factorial is not defined for negative numbers." <<
endl;
    } else {
        for (int i = 1; i <= number; i++) {
            factorial *= i;
        }
        cout << "The factorial of " << number << " is " << factorial
<< endl;
    }
    return 0;
}
```


4.10 QUE NO 3 Fibonacci series

Write a Program in C++ that shows The Fibonacci series

1 1 2 3 5 8 13 Using while loop

4.10.1 Code

```
#include <iostream>

using namespace std;

int main() {
    int number;
    int factorial = 1;
    cout << "Enter a number to calculate its factorial: ";
    cin >> number;
    if (number < 0) {
        cout << "Factorial is not defined for negative numbers." <<
endl;
    } else {
        for (int i = 1; i <= number; i++) {
            factorial *= i;
        }
        cout << "The factorial of " << number << " is " << factorial
<< endl;
    }
    return 0;
}
```


4.11 Que No 4 . Armstrong number

Write a Program in C++ that check the user defined number is Armstrong or not.i.e 153 is an Armstrong number

$$\begin{array}{rccccccc} (1)^3 & + & (5)^3 & + & (3)^3 & & \\ 1 & + & 125 & + & 27 & = & 153 \end{array}$$

4.11.1 Code

```
#include <iostream>
#include <cmath>

using namespace std;

int main() {
    int number, sum = 0, temp, remainder, digits = 0;
    cout << "Enter a number to check if it is Armstrong: ";
    cin >> number;
    temp = number;
    while (temp != 0) {
        temp /= 10;
        digits++;
    }
    temp = number;
    while (temp != 0) {
        remainder = temp % 10;
        sum += pow(remainder, digits);
        temp /= 10;
    }
    if (sum == number) {
        cout << number << " is an Armstrong number." << endl;
    } else {
        cout << number << " is not an Armstrong number." <<
        endl;
    }
    return 0;
}
```


4.12 QUE NO 5 Largest among user defined numbers

Write a Program in C++ that takes Integer values from user, and then find the largest number among all the Integers, and display the largest number on the screen. Use While loop to control the Input, Terminate the Program on entering zero. Find the Largest Integer Using If Statement.

4.12.1 Code

```
#include <iostream>
using namespace std;
int main() {
    int number, largest = 0;
    cout << "Enter integers to find the largest number (enter 0 to stop):\n";
    while (true) {
        cout << "Enter a number: ";
        cin >> number;
        if (number == 0) {
            break;
        }
        if (number > largest) {
            largest = number;
        }
    }
    cout << "The largest number entered is: " << largest << endl;
    return 0;
}
```


4.13 Nested loops Table 12 *12

Write a program in C++ that prints a tables



Starting from 1 12. i.e

1	2	3	4	5	6	7	8	9	10	11	12
2	4	6	8	10	12	14	16	18	20	22	24
3	6	9	12	15	18	21	24	27	30	33	36
4	8	12	16	20	24	28	32	36	40	44	48
5	10	15	20	25	30	35	40	45	50	55	60
6	12	18	24	30	36	42	48	54	60	66	72
7	14	21	28	35	42	49	56	63	70	77	84
8	16	24	32	40	48	56	64	72	80	88	96
9	18	27	36	45	54	63	72	81	90	99	108
10	20	30	40	50	60	70	80	90	100	110	120
11	22	33	44	55	66	77	88	99	110	121	132
12	24	36	48	60	72	84	96	108	120	132	144

4.13.1 Code

```
#include <iostream>
using namespace std;
int main() {
    for (int i = 1; i <= 12; i++) {
        for (int j = 1; j <= 12;
            j++) {
            cout << i * j << "\t";
        }
        cout << endl;
    }
    return 0;
}
```


4.14 Series using nested loops

Develop a code in C++ that generate the following series .Use nested while loop!

Series No.1

1
1 2
1 2 3
1 2 3 4

Series No.2

1
2 2
3 3 3
4 4 4 4

Series No.3

1
2 3
4 5 6
7 8 9 10

4.14.1 Code for Series 1

```

#include <iostream>
using namespace std;
int main() {
    int i = 1, j,
    k;
    cout << "Series No.1\t\tSeries
No.2\t\tSeries No.3\n";
    while (i <= 4) {
        j = 1;
        while (j <= i) {
            cout << j << " ";
            j++;
        }
        cout << "\t\t";
        j = 1;
        while (j <= i) {
            cout << i << " ";
            j++;
        }
        cout << "\t\t";

        k = i * (i - 1) / 2 + 1;
        j = 1;
        while (j <= i) {
            cout << k << " ";
            k++;
            j++;
        }
        cout << endl;
        i++;
    }return 0;
}
```


4.14.2 Code for series 2

```
#include <iostream>
using namespace std;
int main() {
    int i = 1, j;
    while (i <= 4) {
        j = 1;
        while (j <= i) {
            cout << i;
            j++;
        }
        cout << endl;
        i++;
    }
    return 0;
}
```


4.14.3 Code for series 3

```
#include <iostream>
using namespace std;
int main() {
    int num = 1;
    for (int i = 1; i <= 4; i++) {
        for (int j = 1; j <= i; j++) {
            cout << num;
            num++;
        }
        cout << endl;
    }
    return 0;
}
```


Lab Manual # 5

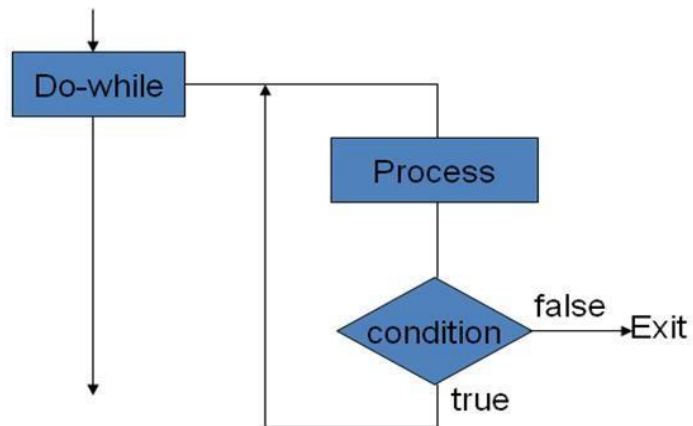
Do-while loop

5.1 do-while loop syntax

```
do  
{  
    statements ;  
}  
while ( condition ) ;
```

5.1.1 Flow Chart of do-while loop

Flow chart for do-while loop



5.2 QUE 1 Calculate the sum of user defined numbers

Develop a program in C++ that take integer type values from user & calculate the sum from 0 to number entered, using do while loop.

Number = 5

$$1+2+3+4+5 = 15$$

5.2.1 Code

```
#include <iostream>
using namespace std;
int main() {
    int number, sum = 0, i = 1;
    cout << "Enter a number: ";
    cin >> number;
    do {
        sum += i;
        i++;
    } while (i <= number);
    cout << "The sum from 1 to " << number << "
is: " << sum << endl;
    return 0;
}
```


5.3 QUE NO 2 Calculate factorial of user defined numbers

Develop a program in C++ that calculate the factorial of a given number. Use for loop to calculate the factorial, & do – while loop to perform the operation as many times as user want.

5.3.1 Code

```
#include <iostream>
using namespace std;
int main() {
    int number;
    char choice;
    do {
        cout << "Enter a number to calculate its factorial: ";
        cin >> number;
        if (number < 0) {
            cout << "Factorial is not defined for negative numbers."
            << endl;
        } else {
            long long factorial = 1;
            for (int i = 1; i <= number; i++) {
                factorial *= i;
            }
            cout << "The factorial of " << number << " is: " <<
            factorial << endl;
        }
        cout << "Do you want to calculate another factorial? (y/n):
        ";
        cin >> choice;
    } while (choice == 'y' || choice == 'Y');

    cout << "Thank you for using the
    factorial calculator!" << endl;
    return 0;
}
```


5.4 Que 3 Calculator

Create the equivalent of a four-function calculator. The program should Ask the user to enter a number, an operator, and another number (10 + 20), using floating point. It should then carry out the specified arithmetical operation: adding, subtracting, multiplying, or dividing the two numbers. Use switch statement to select the operation. Finally, display the Result.

5.4.1 Code

```
#include <iostream>

using namespace std;

int main() {
    float num1, num2, result;
    char operation;
    cout << "Enter first number: ";
    cin >> num1;
    cout << "Enter an operator (+, -, *, /): ";
    cin >> operation;
    cout << "Enter second number: ";
    cin >> num2;
    switch (operation) {
        case '+':
            result = num1 + num2;
            cout << "Result: " << num1 << " + " << num2 << " = "
            << result << endl;
            break;
        case '-':
            result = num1 - num2;
            cout << "Result: " << num1 << " - " << num2 << " = "
            << result << endl;
            break;
        case '*':
            result = num1 * num2;
            cout << "Result: " << num1 << " * " << num2 << " = "
            << result << endl;
            break;
        case '/':
            if (num2 != 0) {
                result = num1 / num2;
                cout << "Result: " << num1 << " / " << num2 << " = "
                << result << endl;
            } else {
                cout << "Error! Division by zero." << endl;
            }
            break;
        default:
            cout << "Error! Invalid operator." << endl;
            break;
    }
    return 0;
}
```


Lab Manual # 6

Function

6.1 Built In Function

```
#include <cmath> // defines the sqrt()
function #include <iostream.h>
```

```
int main()
{ // tests the sqrt() function: for
    (int x=0; x <=9; x++)
        cout << "\t" << x << "\t" << sqrt(x) << endl;
}
```

6.1.1 Out Put

This code will output the square root of numbers from 0 to 9:

```
0 1 2 3 4 5 6 7 8 9
0.00 1.00 1.41 1.73 2.00 2.24 2.45 2.65 2.83 3.00
```

6.2 Write types of Function

Build in functions

User defined functions

Inline functions

Recursive functions

6.3 Write Syntax Of function (Prototype, call & body of function)

```
7 void sort(int arr[], int size) {
8     for (int i = 0; i < size - 1; i++) {
9         // Bubble sort loop
10        for (int j = 0; j < size - 1 - i; j++) {
11            if (arr[j] > arr[j + 1]) {
12                int temp = arr[j];
13                arr[j] = arr[j + 1];
14                arr[j + 1] = temp;
15            }
16        }
17    }
18 }
```


18.1 Que No 1 square () function

Write a program in C++ that has a function of square (). The user should enter the number and the code should display the square of the given number.

18.1.1 Code

```
#include <iostream>

using namespace std;

int square(int num) {
    return num * num;
}

int main() {
    int number;

    cout << "Enter a number: ";
    cin >> number;

    int result = square(number);
    cout << "The square of " << number << " is: " <<
    result << endl;

    return 0;
}
```


18.2 Que No 2 Finding average using function

Get three integers numbers from user, Pass them to function, Add them and find the average in function body, Display the Average in function body

18.2.1 Code

```
#include <iostream>

using namespace std;

int arr[100];
int size = 0;

    cout << "Enter elements to be sorted (enter 0
to stop):\n";
    while (true) {
        cout << "Enter a number: ";
        cin >> arr[size];

        if (arr[size] == 0) {
            break;
        }

        size++;
        if (size >= 100) {
            cout << "Warning: Maximum array size
reached.\n";
            break;
        }
    }
    sort(arr, arr + size);

    cout << "Sorted Array: ";
    for (int i = 0; i < size; i++) {
        cout << arr[i] << " ";
    }
    cout << endl;
}

void calculateAverage(int num1, int num2, int
num3) {
    int sum = num1 + num2 + num3;
    double average = sum / 3.0;
    cout << "The average of " << num1 << ", " <<
num2 << ", and " << num3 << " is: " << average
<< endl;
}

int cube(int n) {
    return n * n * n; // Return the cube of the
number
}

return 0;
```

```
}  
void lab6_program5() {  
    cout << "Hello, this is Program 5 of Lab Manual 6.\n";  
    int num1, num2, num3;  
    cout << "Enter the first number: ";  
    cin >> num1;  
    cout << "Enter the second number: ";  
    cin >> num2;  
    cout << "Enter the third number: ";  
    cin >> num3;  
    calculateAverage(num1, num2, num3);  
}
```


18.3 Que No 3

Write a code that take numbers from user and displays its cube. The Code should reads integers and prints their cubes until the user inputs the sentinel value 0. Each integer read should be passed to the cube() function by the call cube(n). The value returned by the function should replaces the expression cube(n) and then should be passed to the output object cout.

18.3.1 Code

```
#include <iostream>
using namespace std;

int cube(int n) {
    return n * n * n;
}

int main() {
    int num;
    while (true) {
        cout << "Enter a number (0 to exit): ";
        cin >> num;

        if (num == 0) {
            break;
        }

        cout << "Cube of " << num << " is: " <<
        cube(num) << endl;
    }
    return 0;
}
```


18.4 Que No 4 finding area of rectangle

Write a function that finds the area of the rectangle on providing length and width. Get Length & width from user in main() Call the function area() Calculate the length and return the area Display the result in main()

18.4.1 Code

```
#include <iostream>
using namespace std;
int area(int length, int width) {
    return length * width;
}
int main() {
    int length, width, area_result;

    cout << "Enter the length and width of the
rectangle: ";
    cin >> length >> width;

    area_result = area(length, width);

    cout << "The area of the rectangle is: " <<
area_result << endl;
    return 0;
}
```


18.5 Que No 5 Check date program

Get month ,day & year from user in main() Call the function printDate(int, int, int), Put a check in printDate() using if statement (month < 1 || month > 12 || day < 1 || day > 31 || year < 0) & if it violates the rule display —*Must Enter a Valid Date*”, Using switch Statement get the month, Day and year are displayed normally, Termination should be on entering 0 in months

18.5.1 Code

```
#include <iostream>

using namespace std;

int num;
while (true) {
    cout << "Enter a number (enter 0 to stop): ";
    cin >> num;
    if (num == 0) {
        break;
    }
    cout << "The cube of " << num << " is: " << cube(num) <<
endl;
}
}
double area(double length, double width) {
    return length * width;
}
bool isLeapYear(int year) {
    return (year % 4 == 0 && year % 100 != 0) || (year % 400 == 0);
}
void printDate(int month, int day, int year) {
    if (month < 1 || month > 12 || day < 1 || day > 31 || year < 0) {
        cout << "Must enter a valid date.\n";
        return;
    }
    string monthName;
    switch (month) {
        case 1: monthName = "January"; break;
        case 2: monthName = "February"; break;
        case 3: monthName = "March"; break;
        case 4: monthName = "April"; break;
        case 5: monthName = "May"; break;
        case 6: monthName = "June"; break;
        case 7: monthName = "July"; break;
        case 8: monthName = "August"; break;
        case 9: monthName = "September"; break;
        case 10: monthName = "October"; break;
        case 11: monthName = "November"; break;
        case 12: monthName = "December"; break;
        default: monthName = "Invalid"; break;
    }
}
```

```
                                cout << "The entered date is: " << monthName << " " << day << ", " << year << endl;
if (isLeapYear(year)) {
    cout << "The year " << year << " is a leap year.\n";
} else {
    cout << "The year " << year << " is not a leap year.\n";
}
```

18.6 Que No 6 Leap Year program

A leap year is a year in which one extra day (February 29) is added to the regular calendar. Most of us know that the leap years are the years that are divisible by 4. For example, 1992 and 1996 are leap years. Most people, however, do not know that there is an exception to this rule: centennial years are not leap years. For example, 1800 and 1900 are not leap years. Furthermore, there is an exception to the exception: centennial years which are divisible by 400 are leap years. Thus, the year 2000 is a leap year.

- Make a program that full fills the above criteria using functions
- The program should terminate on entering 0
-

The return type should be Boolean i.e `bool isLeapYear(int);`

18.6.1 code

```
#include <iostream>

using namespace std;

bool isLeapYear(int year) {
    if (year % 4 == 0) {
        if (year % 100 == 0) {
            return year % 400 == 0;
        } else {
            return true;
        }
    } else {
        return false;
    }
}

int main() {
    int year;

    cout << "Enter a year: ";
    cin >> year;

    if (isLeapYear(year)) {
        cout << year << " is a leap year." << endl;
    } else {
        cout << year << " is not a leap year." << endl;
    }

    return 0;
}
```


18.7 Que No 7 Finding largest using if else in function

Write a program in C++ that take two numbers from user and find the largest among two using function.

➤ The Conditions are if

(a==b)

"A, B are the same" a = b, (values)

else if (a < b)

"A & B are not same " a != b (values)

"A is less than B " a < b (values)

else

"A is Greater than B " a > b (values)

18.7.1 Code

```
#include <iostream>
using namespace std;
int findLargest(int a, int b) {
    if (a == b) {
        cout << "A and B are the same: " << a << " = "
        << b << endl;
    } else if (a < b) {
        cout << "A and B are not the same" << endl;
        cout << "A is less than B: " << a << " < " << b
        << endl;
    } else {
        cout << "A is greater than B: " << a << " > "
        << b << endl;
    }
    return max(a, b);
}
int main() {
    int num1, num2, largest;

    cout << "Enter two integers: ";
    cin >> num1 >> num2;

    largest = findLargest(num1, num2);

    cout << "The largest number is: " << largest <<
    endl;
    return 0;
}
```


18.8 QueNo 8

Develop a program in C++ that has function printTempOpinion() which prints "Cold" on if the temperature is below 10, "OK" if the temperature is in the range 20 -> 30, "Hot" if the temperature is above 30.

18.8.1 Code

```
#include <iostream>

using namespace std;

void printTempOpinion(int temperature) {
    if (temperature < 10) {
        cout << "Cold" << endl;
    } else if (temperature >= 20 && temperature <=
30) {
        cout << "OK" << endl;
    } else {
        cout << "Hot" << endl;
    }
}

int main() {
    int temperature;

    cout << "Enter the temperature: ";
    cin >> temperature;

    printTempOpinion(temperature);

    return 0;
}
```


18.9 Comparison of functions

Difference between Passing By Value Versus Passing By Reference

By Value	By Reference
Copy of value is passed	Reference to memory location is passed
More efficient for smaller datatypes	Can be more efficient for large data structures
safer	Less safe

18.10 Que No 9 Swap by using function

Develop a program that swap the two values using function by reference

i.e

a = 22.2
b = 44.4

After swap

a = 44.4
b = 22.2

18.10.1 Code

```
#include <iostream>
using namespace std;
void swapValues(int &x, int &y) {
    int temp = x;
    x = y;
    y = temp;
}
int main() {
    int a = 22.2;
    int b = 44.4;

    cout << "Before swap:" << endl;
    cout << "a = " << a << endl;
    cout << "b = " << b << endl;

    swapValues(a, b);

    cout << "After swap:" << endl;
    cout << "a = " << a << endl;
    cout << "b = " << b << endl;

    return 0;
}
```


18.11 Provide an Example of Functions Overloading

```
#include <iostream>

using namespace std;

// Function to add two integers
int add(int a, int b) {
    return a + b;
}

// Function to add two doubles
double add(double a, double b) {
    return a + b;
}

// Function to add three integers
int add(int a, int b, int c) {
    return a + b + c;
}

int main() {
    int x = 5, y = 3, z = 2;
    double p = 2.5, q = 3.7;

    cout << add(x, y) << endl; // Calls add(int, int)
    cout << add(p, q) << endl; // Calls add(double,
double)
    cout << add(x, y, z) << endl; // Calls add(int, int,
int)

    return 0;
}
```


18.12 Que 10 Find Factorial from -1 to 10 using function

```
#include <iostream>

using namespace std;

int factorial(int n) {
    if (n < 0) {
        return -1; // Factorial is not defined for
negative numbers
    } else if (n == 0) {
        return 1;
    } else {
        int fact = 1;
        for (int i = 1; i <= n; i++) {
            fact *= i;
        }
        return fact;
    }
}

int main() {
    for (int i = -1; i <= 10; i++) {
        int result = factorial(i);
        if (result == -1) {
            cout << "Factorial of " << i << " is not
defined." << endl;
        } else {
            cout << "Factorial of " << i << " is " <<
result << endl;
        }
    }
    return 0;
}
```


Lab Manual # 7

1-D Arrays / Strings

7.1 Introduction

They are special kind of data type In C++ each array has

- ✓ name
- ✓ data type
- ✓ size
- ❖ They occupy continuous area of memory

7.2 Que No 1 Displaying age of persons using array

Write a program in C++ that take age of five persons and then just display the age of each person by using arrays.

7.2.1 Code

```
#include <iostream>

using namespace std;

int main() {
    int age[5];

    cout << "Enter the age of 5 persons:\n";
    for (int i = 0; i < 5; i++) {
        cout << "Person " << i + 1 << ": ";
        cin >> age[i];
    }

    cout << "\nAges of the 5 persons:\n";
    for (int i = 0; i < 5; i++) {
        cout << "Person " << i + 1 << ": " << age[i] <<
        " years old\n";
    }
    return 0;
}
```


7.3 Que No 2 Changing values between arrays

Develop a Program that takes array elements from user and then transfer those elements to another array. Size of array will be 10

7.3.1 Code

```
#include <iostream>
using namespace std;
int main() {
    int arr1[10], arr2[10];

    cout << "Enter 10 elements for array 1:\n";
    for (int i = 0; i < 10; i++) {
        cin >> arr1[i];
    }

    // Transfer elements from arr1 to arr2
    for (int i = 0; i < 10; i++) {
        arr2[i] = arr1[i];
    }

    cout << "\nElements of array 2:\n";
    for (int i = 0; i < 10; i++) {
        cout << arr2[i] << " ";
    }

    cout << endl;
    return 0;
}
```


7.4 Que No 3

Develop a program that takes 5 array elements from user. Swap position [2] element with position [4] element.

7.4.1 Code

```
#include <iostream>
using namespace std;
int main() {
    int arr[5];

    cout << "Enter 5 elements:\n";
    for (int i = 0; i < 5; i++) {
        cout << "Element " << i + 1 << ": ";
        cin >> arr[i];
    }

    // Swap elements at positions 2 and 4 (index 1
    and 3)
    int temp = arr[1];
    arr[1] = arr[3];
    arr[3] = temp;

    cout << "\nArray after swapping:\n";

    for (int i = 0; i < 5; i++) {
        cout << arr[i] << " ";
    }

    cout << endl;
    return 0;
}
```


7.5 Que No 4

Write a program to input data into two different arrays and then to add the two arrays and store the result in the third array.

7.5.1 Code

```
#include <iostream>
using namespace std;
int main() {
    int arr1[10], arr2[10], arr3[10];
    int size;

    cout << "Enter the size of the arrays (max 10): ";
    cin >> size;

    cout << "Enter elements for array 1:\n";
    for (int i = 0; i < size; i++) {
        cin >> arr1[i];
    }

    cout << "Enter elements for array 2:\n";
    for (int i = 0; i < size; i++) {
        cin >> arr2[i];
    }

    // Add corresponding elements of arr1 and arr2
    // and store in arr3
    for (int i = 0; i < size; i++) {
        arr3[i] = arr1[i] + arr2[i];
    }

    cout << "\nResultant array:\n";
    for (int i = 0; i < size; i++) {
        cout << arr3[i] << " ";
    }

    cout << endl;

    return 0;
}
```


7.6 Que No 5

Write a program in C++, to input data into an array. Enter a value from the Keyboard and find out the location of the entered value in the array. If the entered number is found in the array, display the message "Number Found—else display —Number Not Found"

7.6.1 Code

```
#include <iostream>
using namespace std;
int main() {
    int arr[10], n, value, found = 0;

    cout << "Enter the size of the array (max 10): ";
    cin >> n;

    cout << "Enter " << n << " elements:\n";
    for (int i = 0; i < n; i++) {
        cin >> arr[i];
    }

    cout << "Enter the value to search: ";
    cin >> value;

    for (int i = 0; i < n; i++) {
        if (arr[i] == value) {
            found = 1;
            cout << "Number found at index " << i <<
endl;
            break;
        }
    }

    if (!found) {
        cout << "Number not found." << endl;
    }

    return 0;
}
```


7.7 Que No 6

Write a program that takes input from user and checks if the word is Palindrome or Not

7.7.1 Code

```
#include <iostream>
#include <string>

using namespace std;

bool isPalindrome(string str) {
    int n = str.length();
    for (int i = 0; i < n / 2; i++) {
        if (str[i] != str[n - i - 1]) {
            return false;
        }
    }
    return true;
}

int main() {
    string word;

    cout << "Enter a word: ";
    cin >> word;

    if (isPalindrome(word)) {
        cout << word << " is a palindrome." << endl;
    } else {
        cout << word << " is not a palindrome." <<
endl;
    }

    return 0;
}
```


7.8 Que No7

Find the Average of 10 numbers passed to an array, using function. The Array should be controlled by while loop.

7.8.1 Code

```
#include <iostream>
using namespace std;
double calculateAverage(int arr[], int size) {
    int sum = 0;
    for (int i = 0; i < size; i++) {
        sum += arr[i];
    }
    return (double)sum / size;
}
int main() {
    int arr[10];
    int i = 0;

    cout << "Enter 10 numbers: ";

    while (i < 10) {
        cin >> arr[i];
        i++;
    }

    double average = calculateAverage(arr, 10);

    cout << "Average of the numbers: " << average
    << endl;

    return 0;
}
```


7.9 Que No 8

Write a Program in C++ that do the bubble sorting in Array. Hint

- ✓ Array Elements are entered by user in —main functionll, the elements are user Defined. Terminate on “zero”
- ✓ Make a separate function “sort()” to do the bubble sort.
- ✓ Pass array elements to function “sort()”.
- ✓ Function should sort the array elements.
- ✓ Display the sorted array elements in main.

7.9.1 Code

```
#include <iostream>

using namespace std;

void sort(int arr[], int n) {
    for (int i = 0; i < n - 1; i++) {
        for (int j = 0; j < n - i - 1; j++) {
            if (arr[j] > arr[j + 1]) {
                swap(arr[j], arr[j + 1]);
            }
        }
    }
}

int main() {
    int arr[100], n = 0;

    cout << "Enter array elements (terminate with 0): ";
    while (n < 100 && cin >> arr[n] && arr[n] != 0)
    {
        n++;
    }

    sort(arr, n);

    cout << "Sorted array: ";
    for (int i = 0; i < n; i++) {
        cout << arr[i] << " ";
    }
    cout << endl;

    return 0;
}
```


7.10 Strings

A string constant is a one-dimensional array of characters terminated by a null (`'\0'`).
For example,

```
char name[ ] = { 'H', 'A', 'E', 'S', 'L', 'E', 'R', '\0' } ;
```

7.10.1 Example of Strings

```
/* Program to demonstrate printing of a string */
main( )
{
char name[ ] = "Klinsman" ;
int i = 0 ;
while ( i <= 7 )
{
    cout<<name[i];
    i++ ;
}
}
```

```
/* Program to demonstrate printing of a string */
main( )
{
char name[ ] = "Klinsman" ;
int i = 0 ;
while (name[i] !='\0')
{
    cout<<name[i];
    i++ ;
}
}
```

7.10.2 Que No 9 Get the name from user

Write a program that gets the name from user and then print back on the screen.

```
#include <iostream>
#include <string>

using namespace std;

int main() {
    string name;

    cout << "Enter your name: ";
    getline(cin, name);

    cout << "Hello, " << name << "!" << endl;

    return 0;
}
```

7.11 The String I/O Function gets() & puts()

cin & cout are not so versatile because they ignore characters after white space, so to overcome this problem C/C++ uses puts & gets function from library file <stdio.h>

7.12 The String I/O Function gets() & puts()

Write a program that gets the name from user and then print back on the screen using gets and puts functions.

```
printf("Enter your name: ");  
gets(name);  
printf("Hello, %s!\n", name);  
return 0;  
}
```

```
#include <stdio.h>

int main() {
    char name[100];

    // Input the string
    printf("Enter your name: ");
    gets(name); // Unsafe, for educational purposes only

    // Output the string
    printf("Hello, ");
    puts(name); // Prints the string with a newline

    return 0;
}
```

7.13 strcpy & strcmp

```
#include <iostream>
#include <cstring>
using namespace std;

int main() {
    char str1[50], str2[50];

    // Using strcpy
    strcpy(str1, "Hello");
    cout << "String 1: " << str1 << endl;

    // Input for string 2
    cout << "Enter another string: ";
    cin.getline(str2, 50);

    // Using strcmp
    int result = strcmp(str1, str2);
    if (result == 0)
        cout << "Both strings are equal." << endl;
    else if (result > 0)
        cout << "String 1 is greater." << endl;
    else
        cout << "String 2 is greater." << endl;

    return 0;
}
```

7.13.1 strcpy & strcmp example

--

7.14 Que No 9 String Deletion

Write a program in C++ that take string from user and then enter the position to delete. Make a separate function for string deletion

```
#include <iostream>
#include <string>
using namespace std;

// Function to delete a character at a specified position
void deleteCharacter(string &str, int position) {
    if (position < 0 || position >= str.length()) {
        cout << "Invalid position!" << endl;
        return;
    }
    str.erase(position, 1); // Erase one character at the specified
    position
}

int main() {
    string userInput;
    int position;

    // Input string from the user
    cout << "Enter a string: ";
    getline(cin, userInput);

    // Input the position to delete
    cout << "Enter the position of the character to delete (0-based
    index): ";
```

```
        cin >> position;

        // Call the function to delete the character
        deleteCharacter(userInput, position);

        // Output the modified string
        cout << "Modified string: " << userInput << endl;

        return 0;
}
```


Lab Manual # 8

2-D Arrays

8.1 Que No 1 Shows sale of Pharmaceutical distribution company

A Distributor of a Pharmaceutical Company has 4 Districts, for supply the medicine. He requires a program that can display the sales of all his Districts. Write a Program in C++ Using two Dimensional Array that shows the Following Output. The Program should display the Sale, Districts wise and up to Months i.e

8.1.1 Code

```
#include <iostream>
#include <iomanip>

using namespace std;

int main() {
    // Declare a 2D array to store sales data
    int sales[4][12] = {
        {1000, 1200, 1500, 1800, 2000, 2200, 2500,
         2800, 3000, 3200, 3500, 4000},
        {800, 900, 1100, 1300, 1500, 1700, 1900, 2100,
         2300, 2500, 2700, 3000},
        {1200, 1400, 1600, 1800, 2000, 2200, 2400,
         2600, 2800, 3000, 3200, 3500},
        {900, 1000, 1200, 1400, 1600, 1800, 2000,
         2200, 2400, 2600, 2800, 3000}
    };

    // Array of month names
    string months[] = {"Jan", "Feb", "Mar", "Apr",
                       "May", "Jun", "Jul", "Aug", "Sep", "Oct", "Nov",
                       "Dec"};

    // Print the header
    cout << setw(10) << "District" << setw(10) <<
    "Jan" << setw(10) << "Feb" << setw(10) << "Mar"
    << setw(10) << "Apr" << setw(10) << "May" <<
    setw(10) << "Jun" << setw(10) << "Jul" << setw(10)
    << "Aug" << setw(10) << "Sep" << setw(10) <<
    "Oct" << setw(10) << "Nov" << setw(10) << "Dec"
    << endl;

    // Print the sales data for each district
    for (int i = 0; i < 4; i++) {
        cout << setw(10) << "District " << i + 1;
        for (int j = 0; j < 12; j++) {
            cout << setw(10) << sales[i][j];
        }
        cout << endl;
    }

    return 0;
}
```


8.2 Que no 2 Adding two matrices

Write a program in C++ that take two matrices and then Add them After inserting two matrices first display the both matrices and then add them and show the result. i.e

8.2.1 Code

```
#include <iostream>

using namespace std;

int main() {
    int m, n, p, q;

    // Input dimensions of the first matrix
    cout << "Enter the dimensions of the first matrix\n(m x n): ";
    cin >> m >> n;

    // Input dimensions of the second matrix
    cout << "Enter the dimensions of the second\nmatrix (p x q): ";
    cin >> p >> q;

    // Check if matrices are compatible for addition
    if (m != p || n != q) {
        cout << "Matrices cannot be added.\nDimensions must be the same." << endl;
        return 1;
    }

    // Declare matrices
    int A[m][n], B[m][n], C[m][n];

    // Input elements of the first matrix
    cout << "Enter elements of the first matrix:" << endl;
    for (int i = 0; i < m; i++) {
        for (int j = 0; j < n; j++) {
            cin >> A[i][j];
        }
    }

    // Input elements of the second matrix
    cout << "Enter elements of the second matrix:" << endl;
    for (int i = 0; i < m; i++) {
        for (int j = 0; j < n; j++) {
            cin >> B[i][j];
        }
    }

    // Display the first matrix
    cout << "First matrix:" << endl;
    for (int i = 0; i < m; i++) {
        for (int j = 0; j < n; j++) {
            cout << A[i][j] << " ";
        }
        cout << endl;
    }

    // Display the second matrix
    cout << "Second matrix:" << endl;
    for (int i = 0; i < m; i++) {
        for (int j = 0; j < n; j++) {
            cout << B[i][j] << " ";
        }
        cout << endl;
    }
}
```

```

    }
    // Add the matrices
    for (int i = 0; i < m; i++) {
        for (int j = 0; j < n; j++) {
            C[i][j] = A[i][j] + B[i][j];
        }
    }

    // Display the sum matrix
    cout << "Sum of matrices:" << endl;
    for (int i = 0; i < m; i++) {
        for (int j = 0; j < n; j++) {
            cout << C[i][j] << " ";
        }
        cout << endl;
    }

    return 0;
}

```


8.4 Que no 3 Printing matrix in reserve.

Enter the values in a matrix and print it in reverse Column order

8.4.1 Code

```
#include <iostream>

using namespace std;

int main() {
    int rows, cols;

    cout << "Enter the number of rows: ";
    cin >> rows;

    cout << "Enter the number of columns: ";
    cin >> cols;

    int matrix[rows][cols];

    cout << "Enter the elements of the matrix:\n";
    for (int i = 0; i < rows; i++) {
        for (int j = 0; j < cols; j++) {
            cin >> matrix[i][j];
        }
    }

    cout << "Matrix in reverse column order:\n";
    for (int j = cols - 1; j >= 0; j--) {
        for (int i = 0; i < rows; i++) {
            cout << matrix[i][j] << " ";
        }
        cout << endl;
    }

    return 0;
}
```


8.5 Que no 4 Transpose of a Matrix

Write a Program in C++ that Display the Transpose of a Matrix.

8.5.1 Code

```
#include <iostream>

using namespace std;

int main() {
    int rows, cols;

    cout << "Enter the number of rows: ";
    cin >> rows;

    cout << "Enter the number of columns: ";
    cin >> cols;

    int matrix[rows][cols];

    cout << "Enter the elements of the matrix:\n";
    for (int i = 0; i < rows; i++) {
        for (int j = 0; j < cols; j++) {
            cin >> matrix[i][j];
        }
    }

    cout << "Original Matrix:\n";
    for (int i = 0; i < rows; i++) {
        for (int j = 0; j < cols; j++) {
            cout << matrix[i][j] << " ";
        }
        cout << endl;
    }

    cout << "Transpose Matrix:\n";
    for (int i = 0; i < cols; i++) {
        for (int j = 0; j < rows; j++) {
            cout << matrix[j][i] << " ";
        }
        cout << endl;
    }

    return 0;
}
```


8.6 Que no 5 Agent Program

Write a program in C++, which take Agent code (123,258,..) and Traveling expense (Rs = 5000, 6000,...) of the agent. Find the agent who had spent most money in all, Display the agent code and amount after searching in 2 D Array.

8.6.1 Code

```
#include <iostream>
using namespace std;
int main() {
    int n;

    cout << "Enter the number of agents: ";
    cin >> n;

    int agent_code[n], travel_expense[n];

    cout << "Enter agent codes and travel
expenses:\n";
    for (int i = 0; i < n; i++) {
        cout << "Agent " << i + 1 << ": " << endl;
        cout << "  Code: ";
        cin >> agent_code[i];
        cout << "  Expense: ";
        cin >> travel_expense[i];
    }

    int max_expense = travel_expense[0];
    int max_index = 0;

    for (int i = 1; i < n; i++) {
        if (travel_expense[i] > max_expense) {
            max_expense = travel_expense[i];
            max_index = i;
        }
    }

    cout << "\nAgent with the highest expense:\n";
    cout << "  Code: " << agent_code[max_index] <<
endl;
    cout << "  Expense: Rs." << max_expense <<
endl;

    return 0;
}
```


Lab Manual # 9

Structures

9.1 Introduction

- A Structure is a collection of simple variables. The Variables in a structure can be of different types. Some can be int, some can be float, and so on.
- The data items in a structure are called the members of the structure.
- The structure is a kind of blue print specifying what information is necessary for a single part.

9.2 Structure Example

```
struct part
{
    int modelnumber;
    int partnumber;
    float cost;
};
int main ()
{
    part part1, part2;
    part1.modelnumber =
    1111; part1.partnumber =
    111; part1.cost = 111.11;

    part2.modelnumber = 222;
    part2.partnumber = 2222;
    part2.cost = 222.222;
    cout<<"\nModel of Part1 = "<<part1.modelnumber;
    cout<<"\nPart of part 1 = "<<part1.partnumber;
    cout<<"\nCost of part1 = "<<part1.cost<<endl;

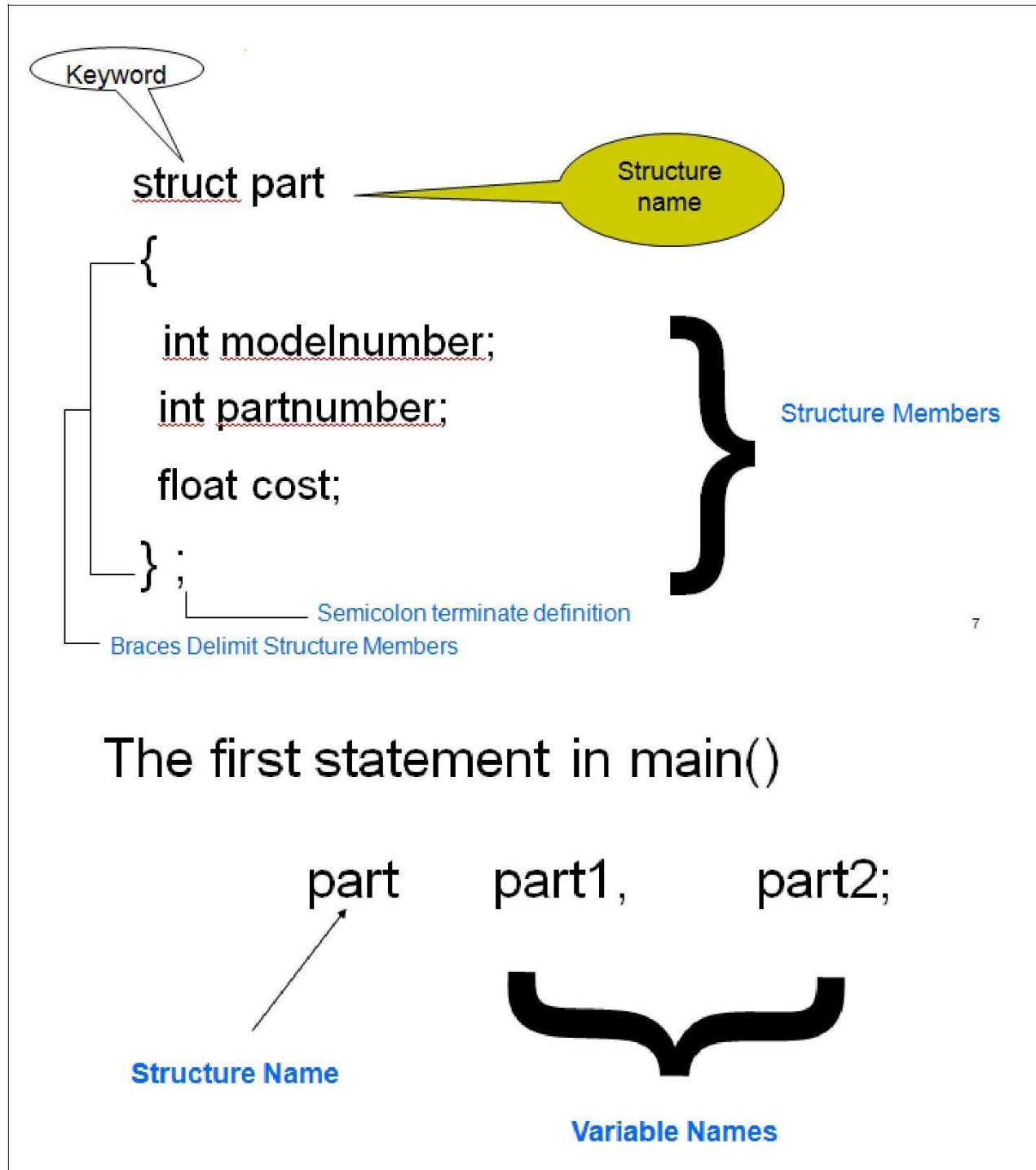
    cout<<"\nModel of part2 = "<<part2.modelnumber;
    cout<<"\nPart of part2 = "<<part2.partnumber;
    cout<<"\nCost of part2 = "<<part2.cost<<endl;
    return 0;
}
```

9.2.1 Output

```
Model of Part1 = 1111
Part of part 1 = 111
Cost of part1 = 111.11
Model of part2 = 222
Part of part2 = 2222
Cost of part2 = 222.222
```

9.2.2 Description

The structure definition serves only as a blueprint for the creation of variables of type part. It does not itself create any structure variables; that is, it does not set aside any space in memory or even name any variables. This is unlike the definition of a simple variable, which does set aside memory.



9.3 Que No 1 Area of rooms

Write a program in C++ that shows the area of 3 room's. Using Structure namely "distance". Take input of feet & inches from user for variable d1 (feet & inches), assign variable d2 = {10, 5.25} values. Now add feet and inches of d1 & d2 and store in d3.

Display d1 (feet & inches) d2 (feet & inches) d3 (feet & inches) separately. Put Condition if d1 & d2 inches increase by 12 it become a foot.

9.3.1 Code

```
10 struct distance {
11     int feet;
12     float inches;
13 };
14
15 {
16     distance d1, d2, d3;
17     cout << "Enter distance d1 (feet): ";
18     cin >> d1.feet;
19     cout << "Enter distance d1 (inches): ";
20     cin >> d1.inches;
21     d2.feet = 10;
22     d2.inches = 5.25;
23     d3.feet = d1.feet + d2.feet;
24     d3.inches = d1.inches + d2.inches;
25     if (d3.inches >= 12) {
26         d3.feet += static_cast<int>(d3.inches / 12);
27         d3.inches = fmod(d3.inches, 12);
28     }
29     cout << "\nDistance d1: " << d1.feet << " feet, " << d1.inches << "
    inches";
30     cout << "\nDistance d2: " << d2.feet << " feet, " << d2.inches << "
    inches";
31     cout << "\nDistance d3 (sum): " << d3.feet << " feet, " << d3.inches <<
    " inches";
32 }
33
```


33.1 Que No 2 Phone Number Program

A phone number, such as 212- 767-8900, can be thought of as having three parts: the area code (212), the exchange (767), and the number (8900). Write a program that uses a structure to store these three parts of a phone number separately. Call the structure phone. Create two structures variables of type phone. Initialize one, and have the user input a number for the other one. Then display both numbers. The interchange might look like this:

33.1.1 Code

```
34
35 #include <iostream>
36 using namespace std;
37
38 // Define a structure for the phone number
39 struct Phone {
40     int areaCode;
41     int exchange;
42     int number;
43 };
44
45 // Function to initialize and display phone numbers
46 void handlePhoneNumbers() {
47     Phone phone1 = {212, 767, 8900}; // Initialize the first phone number
48     Phone phone2; // Create a second phone number variable
49
50     // Prompt the user to input the second phone number
51     cout << "Enter your phone number details:" << endl;
52     cout << "Area code: ";
53     cin >> phone2.areaCode;
54     cout << "Exchange: ";
55     cin >> phone2.exchange;
56     cout << "Number: ";
57     cin >> phone2.number;
58
59     // Display both phone numbers
60     cout << "\nPhone number 1: (" << phone1.areaCode << ") " << phone1.exchange << "-"
    << phone1.number << endl;
61     cout << "Phone number 2: (" << phone2.areaCode << ") " << phone2.exchange << "-"
    << phone2.number << endl;
62 }
63
64 // Entry point (main) function, calling handlePhoneNumbers
65 void main_body() {
66     handlePhoneNumbers(); // Call the function to handle phone numbers
67 }
```

67.1 Que No 3 Employee record

Create a structure called emp that contains three members,
int id, char name[100], float sal.

Ask the user to fill in data for three employees and then display information for each employee.

Hint

- Variable of struct emp will be array
- Use while / for loop to control array

67.1.1 Code

```
68  struct emp {
69      int id;
70      char name[100];
71      float sal;
72  };
73  emp employees[3];
74
75  for (int i = 0; i < 3; i++) {
76      cout << "Enter details for employee " << i + 1 << ":\n";
77      cout << "ID: ";
78      cin >> employees[i].id;
79      cin.ignore();
80      cout << "Name: ";
81      cin.getline(employees[i].name, 100);
82      cout << "Salary: ";
83      cin >> employees[i].sal;
84      cout << endl;
85  }
86
87  cout << "Employee details:\n";
88  for (int i = 0; i < 3; i++) {
89      cout << "\nEmployee " << i + 1 << ":\n";
90      cout << "ID: " << employees[i].id << endl;
91      cout << "Name: " << employees[i].name << endl;
92      cout << "Salary: " << employees[i].sal << endl;
93  }
94
```

9.5 Que No 4 Memory size of a structure

Write a program using —sizeof() function that calculate the size of structure

9.5.1 Code

```
10 #include <iostream>
11 using namespace std;
12
13 // Define a structure
14 struct Phone {
15     int areaCode;
16     int exchange;
17     int number;
18 };
19
20 void calculateSizeOfStructure() {
21     // Create a Phone structure variable
22     Phone phone;
23
24     // Display the size of the structure
25     cout << "Size of the Phone structure: " << sizeof(phone) << " bytes" << endl;
26 }
27
28 void main_body() {
29     calculateSizeOfStructure(); // Call the function to calculate the size of the structure
30 }
```

30.5 Que No 5 Average Age Program

Write a program to calculate the average age and average GPA of a class having 10 students. Also determine the grade of the class and the student with max GPA. Use a student structure and manipulate it to get the desired result.

30.5.1 Code

```
31     struct Student {
32         string name;
33         int age;
34         float gpa;
35     };
36     Student students[10];
37     float totalAge = 0, totalGPA = 0;
38     int maxGPAIndex = 0;
39
40     for (int i = 0; i < 10; i++) {
41         cout << "Enter details for student " << i + 1 << ":\n";
42         cout << "Name: ";
43         cin.ignore();
44         getline(cin, students[i].name);
45         cout << "Age: ";
46         cin >> students[i].age;
47         cout << "GPA: ";
48         cin >> students[i].gpa;
49
50         totalAge += students[i].age;
51         totalGPA += students[i].gpa;
52
53         if (students[i].gpa > students[maxGPAIndex].gpa) {
54             maxGPAIndex = i;
55         }
56     }
57
58     float avgAge = totalAge / 10;
59     float avgGPA = totalGPA / 10;
60
61     char classGrade;
62     if (avgGPA >= 3.5) {
63         classGrade = 'A';
64     } else if (avgGPA >= 3.0) {
65         classGrade = 'B';
66     } else if (avgGPA >= 2.5) {
67         classGrade = 'C';
68     } else {
69         classGrade = 'D';
70     }
71
```

```
72     cout << "\nClass Summary:\n";
73     cout << "Average Age: " << avgAge << endl;
74     cout << "Average GPA: " << avgGPA << endl;
75     cout << "Class Grade: " << classGrade << endl;
76     cout << "Student with the highest GPA:\n";
77     cout << "Name: " << students[maxGPAIndex].name << endl;
78     cout << "Age: " << students[maxGPAIndex].age << endl;
79     cout << "GPA: " << students[maxGPAIndex].gpa << endl;
80 }
81
82
```

82.5 Que No 6 Nested Structure

Write a program that contains nested structure

82.5.1 Code

```
83 struct Student {
84     string name;
85     int age;
86     float gpa;
87 };
88 Student students[10];
89 float totalAge = 0, totalGPA = 0;
90 int maxGPAIndex = 0;
91
92 for (int i = 0; i < 10; i++) {
93     cout << "Enter details for student " << i + 1 << ":\n";
94     cout << "Name: ";
95     cin.ignore();
96     getline(cin, students[i].name);
97     cout << "Age: ";
98     cin >> students[i].age;
99     cout << "GPA: ";
100    cin >> students[i].gpa;
101
102    totalAge += students[i].age;
103    totalGPA += students[i].gpa;
104
105    if (students[i].gpa > students[maxGPAIndex].gpa) {
106        maxGPAIndex = i;
107    }
108 }
109
110 float avgAge = totalAge / 10;
111 float avgGPA = totalGPA / 10;
112
113 char classGrade;
114 if (avgGPA >= 3.5) {
115     classGrade = 'A';
116 } else if (avgGPA >= 3.0) {
117     classGrade = 'B';
118 } else if (avgGPA >= 2.5) {
119     classGrade = 'C';
120 } else {
121     classGrade = 'D';
122 }
123
124 cout << "\nClass Summary:\n";
125 cout << "Average Age: " << avgAge << endl;
```

```
126     cout << "Average GPA: " << avgGPA << endl;
127     cout << "Class Grade: " << classGrade << endl;
128     cout << "Student with the highest GPA:\n";
129     cout << "Name: " << students[maxGPAIndex].name << endl;
130     cout << "Age: " << students[maxGPAIndex].age << endl;
131     cout << "GPA: " << students[maxGPAIndex].gpa << endl;
132 }
133
134
```

134.5 Que No 7 Access of structure data members with pointer to structure

```
135 struct Academic {
136     float gpa;
137     string grade;
138 };
139
140 struct Personal {
141     string name;
142     int age;
143     string address;
144 };
145
146 struct Student {
147     Personal personalDetails;
148     Academic academicDetails;
149 };
150
151 {
152     Student students[2];
153
154     for (int i = 0; i < 2; i++) {
155         cout << "Enter details for student " << i + 1 << ":\n";
156         cout << "Name: ";
157         cin.ignore();
158         getline(cin, students[i].personalDetails.name);
159         cout << "Age: ";
160         cin >> students[i].personalDetails.age;
161         cin.ignore();
162         cout << "Address: ";
163         getline(cin, students[i].personalDetails.address);
164         cout << "GPA: ";
165         cin >> students[i].academicDetails.gpa;
166         cin.ignore();
167         cout << "Grade: ";
168         getline(cin, students[i].academicDetails.grade);
169         cout << endl;
170     }
171     cout << "Student Details:\n";
172     for (int i = 0; i < 2; i++) {
173         cout << "\nStudent " << i + 1 << ":\n";
174         cout << "Name: " << students[i].personalDetails.name << endl;
175         cout << "Age: " << students[i].personalDetails.age << endl;
176         cout << "Address: " << students[i].personalDetails.address << endl;
177         cout << "GPA: " << students[i].academicDetails.gpa << endl;
178         cout << "Grade: " << students[i].academicDetails.grade << endl;
179     }
180 }
```


181 }

182

Lab Manual # 10

Pointers

10.1 Introduction to Pointers

```
int *myptr ;  
    myptr is pointer to an integer
```

10.1.1 Example of Pointers

```
#include<iostream.h>  
void main ()  
{  
    int var1 = 10;  
    int var2 = 20;  
    int var3 = 30;  
    cout<<&var1<<endl  
        <<&var2<<endl;  
    int *ptr;  
    ptr = &var1;  
    cout<<ptr<<endl;  
    ptr = &var2;  
    cout<<ptr<<endl;  
}
```

10.2 Pointer To Arrays

```
#include<iostream.h>  
void main ()  
{  
    int array[5] = {31,54,77,52,93};  
    for(int j =0; j<5; j++)  
    {  
        cout<<array[j]<<endl;  
    }  
}
```

```
#include<iostream.h>  
void main ()  
{  
    int array[5] = {31,54,77,52,93};  
    int* ptr;  
    ptr = array;  
    for(int j =0; j<5; j++)  
    {  
        cout<<*(ptr++)<<endl;  
    }  
}
```

10.2 Que No 1 Print the values from array

Write a program that prints the values from an array using pointer variable. The array is given below

```
int y [ 10 ]= {6,2,3,12};
```

```
int array[5] = {31, 54, 77, 52, 93};
cout << "Displaying array elements using indexing:\n";
for (int j = 0; j < 5; j++) {
    cout << array[j] << endl;
}
cout << "\nDisplaying array elements using a pointer:\n";
int* ptr = array;
for (int j = 0; j < 5; j++) {
    cout << *(ptr++) << endl;
}
cout << "\nAnother example with a second array:\n";
int y[10] = {6, 2, 3, 12};
ptr = y;
cout << "Array elements printed using pointer:\n";
for (int i = 0; i < 10; i++) {
    cout << *(ptr + i) << " ";
}
cout << endl;
}
```

10.3 Que No 2 Print the values and memory address from an array

Write a program that prints the values from an array using pointer variable. The array is given below

```
int y [ 10 ]= {6,2,3,12};
```

```
int y[10] = {6, 2, 3, 12};
int* ptr = y;
cout << "Array values and their memory addresses:\n";
for (int i = 0; i < 10; i++) {
    cout << "Value: " << *(ptr + i) << ", Address: " << (ptr + i)
<< endl;
}
```

10.4 Pointer Arithmetic

```
int x =10 ;  
int *yptr ;  
yptr = &x ;  
*yptr += 3 ;  
yptr += 3 ;
```

10.4.1 Example of Pointer arithmetic

```
long* pnumber = NULL;  
long number1 = 10, number2 = 20;  
  
pnumber = &number1;  
*pnumber += 2;  
  
cout<<"\nnumber1 = "<<number1  
    <<" &number = "<<pnumber;  
  
pnumber = &number2;  
number1 = *pnumber *4;  
  
cout<<"\nnumber1 = "<<number1 <<"  
    pnumber = "<<pnumber  
    <<"pnumber = "<<*pnumber;
```

10.4.2 Output

```
number1    =    12    &number    =  
0x0012FF78 number1 = 80 pnumber =  
0x0012FF74 pnumber = 20  
Press any key to continue
```

10.5 Que No 3 Accessing values by Arithmetic operator

Write a program that displays the values using pointer variable from an array given below using Arithmetic Increment operator .

```
int y[5]={22,33,44,55,66};
```

10.5.1 Code

```
11     long* pnumber = NULL;
12     long number1 = 10, number2 = 20;
13     pnumber = &number1;
14     *pnumber += 2;
15     cout << "\nnumber1 = " << number1 << " &pnumber = " << pnumber <<
endl;
16     pnumber = &number2;
17     number1 = *pnumber * 4;
18     cout << "\nnumber1 = " << number1 << " pnumber = " << pnumber << "
*pnumber = " << *pnumber << endl;
19
```

19.2 Que No 4 Moving in array through pointers

Write a program that display only 6th element of an array given below using pointers.

```
int y [10] = {11,22,33, 44,55,66,77,88,99,110}
```

19.2.1 Code

```
20  cout << "Hello, this is Program 6 of Lab Manual 10.\n";
21      int y[10] = {11, 22, 33, 44, 55, 66, 77, 88, 99, 110};
22      int* ptr = y;
23      cout << "The 6th element of the array is: " << *(ptr + 5) << endl;
24
```

24.2 Pointer Comparison

```
if ( y1 > y2 )
if ( y1 >= y2 )
if ( y1 == y2 )
```

```
if ( *y1 > *y2 )
```


24.2.1 Pointer Comparison Example

```
int y [10]={11,22,33,44,55,66,77,88,99,110} ;

int *y1, *y2;
y1= &y[0];
y2= &y[3];

cout <<"\n Y1= "<<*y1;
cout <<"\n Y2= "<<*y2;

if (*y1 < *y2)
    cout<<"\nY1 is Smaller"<<endl;
else
    cout<<"\nY2 is smaller"<<endl;
```

24.3 Que No 4 Question Max

Write a program that take two numbers an input from user. Find the maximum from both of them using the dereference pointer comparison

24.3.1 Code

```
25     long* pnumber = NULL;
26     long number1 = 10, number2 = 20;
27     pnumber = &number1;
28     *pnumber += 2;
29     cout << "\nnumber1 = " << number1 << " &pnumber = " << pnumber <<
endl;
30     pnumber = &number2;
31     number1 = *pnumber * 4;
32     cout << "\nnumber1 = " << number1 << " pnumber = " << pnumber << "
*pnumber = " << *pnumber << endl;
33 }
34
35
```

35.2 Pointer to functions

```
main( )
{
int a = 10, b = 20 ;
swapv ( a, b ) ;
cout<<"\na ="<<a<<" b= "<<b;
}
swapv ( int x, int y )
{
int t ; t
= x ; x
= y ; y
= t ;
cout<<"\nx = "<<x<<" y = "<<y;
}
```

The above given code is swapping the values without pointers.

35.2.1 Que No 5 Swap the same values using pointers.

```
36  long* pnumber = NULL;
37  long number1 = 10, number2 = 20;
38  pnumber = &number1;
39  *pnumber += 2;
40  cout << "\nnumber1 = " << number1 << " &pnumber = " << pnumber <<
    endl;
41  pnumber = &number2;
42  number1 = *pnumber * 4;
43  cout << "\nnumber1 = " << number1 << " pnumber = " << pnumber << "
    *pnumber = " << *pnumber << endl;
44  }
45
```

45.2 Que No 6 Returning more than one values from a function

Write a program that gets the radius from user, pass radius to a function `areaperi()` and function `areaperi()` returns —areall and —perimeterll by reference

Using a call by reference intelligently we can make a function return more than one value at a time, which is not possible ordinarily.

45.2.1 Code

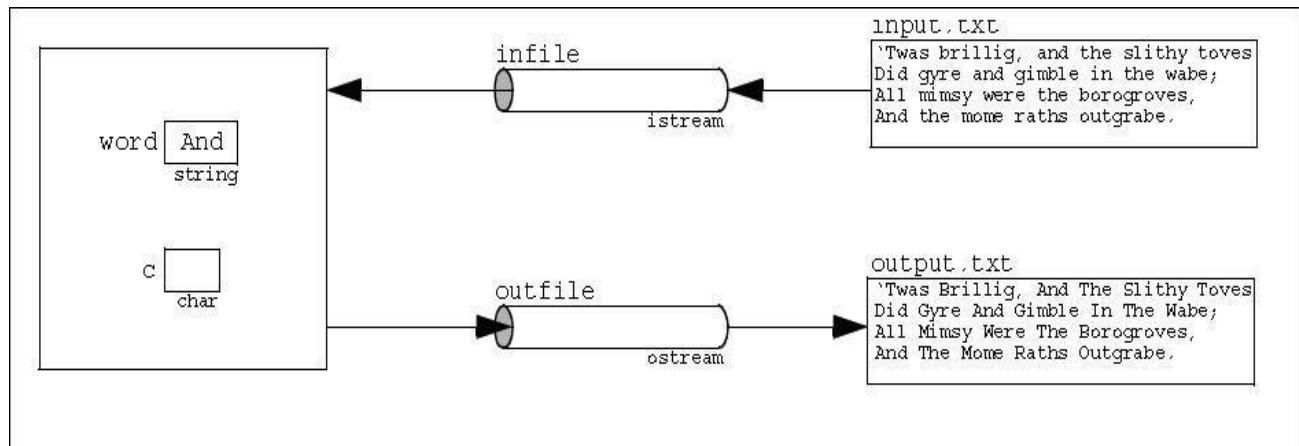
```
46  #include <iostream>
47  #include <cmath> // For M_PI (value of PI)
48
49  using namespace std;
50
51  // Function to calculate area and perimeter by reference
52  void areaperi(double radius, double &area, double &perimeter) {
53      // Calculate the area and perimeter of the circle
54      area = M_PI * radius * radius;      // Area =  $\pi * r^2$ 
55      perimeter = 2 * M_PI * radius;      // Perimeter (Circumference) =  $2 * \pi * r$ 
56  }
57
58  int main() {
59      double radius;
60      double area, perimeter;
61
62      // Input the radius from the user
63      cout << "Enter the radius of the circle: ";
64      cin >> radius;
65
66      // Call the function and pass area and perimeter by reference
67      areaperi(radius, area, perimeter);
68
69      // Output the area and perimeter
70      cout << "Area of the circle: " << area << endl;
71      cout << "Perimeter (Circumference) of the circle: " << perimeter << endl;
72
73      return 0;
74  }
```

Lab Manual # 11

Files

11.1 Introduction to Files

File processing in C++ is very similar to ordinary interactive input and output because the same kinds of stream objects are used. Input from a file is managed by an ifstream object the same way that input from the keyboard is managed by the istream object cin. Similarly, output to a file is managed by an ofstream object the same way that output to the monitor or printer is managed by the ostream object cout. The only difference is that ifstream and ofstream objects have to be declared explicitly and initialized with the external name of the file which they manage. You also have to #include the <fstream> header file (or <fstream.h> in pre-Standard C++) that defines these classes



Notice that the program has four objects: an ifstream object named infile, an ofstream object named outfile, a string object named word, and a char object named c.

The advantage of using external files instead of command line redirection is that there is no limit to the number of different files that you can use in the same program.

11.2 Files of C

```
//Program that create / open a
file #include<stdio.h>
#include<conio.h>
void main ()
{
    FILE *fptr;
    char ch;
    fptr = fopen("first.txt","w");
    while((ch =getche()) !=
        '\r') putc(ch,fptr);
    fclose(fptr);
}
```

11.3 Files in C++

```
include <fstream>
using namespace std;
int main ()
{
    ofstream myfile;
    myfile.open ("first.txt");

    myfile << "This is my first file
    \n"; myfile.close();
    return 0;
}
```

This Code will creates a file —first.txtll and insert a sentence into it in the same way we are used to do with cout<< but using the myfile stream instead

11.3.1 Open a file

Open (filename, mode);

■ Mode is optional parameter with a combination of the following flags

ios::in	Open for input operations.
ios::out	Open for output operations.
ios::binary	Open in binary mode. Set the initial position at the end of the file.
ios::ate	If this flag is not set to any value, the initial position is the beginning of the file.
ios::app	All output operations are performed at the end of the file, appending the content to the current content of the file. This flag can only be used in streams open for output-only operations.
ios::trunc	If the file opened for output operations already existed before, its previous content is deleted and replaced by the new one.

All these flags can be combined using the bitwise operator OR (|). For example, if we want to open the file example.bin in binary mode to add data we could do it by the following call to member function open(): `ofstream myfile;myfile.open ("example.bin", ios::out | ios::app | ios::binary);`

11.4 Que No 1 Create a text file

Write a program that create a text file —example.txtll. Open that file and write two line in it —This is a line and This is another linell and close the file. If the file is unable to create or open then show a message —ERROR Unable to open a filell.

11.4.1 Code

11.5 Que No 2 Read from the file

Write a program that read the text from file —example.txtll.

If the file is unable to open then show a message —ERROR Unable to open a filell.

11.5.1 Code

11.6 Que No 3 Write data through variable

Write a program that create a text file —fdata.txtll. Open that file and write the values through variables as under. After successful writing show a message —File written successfullyll .

```
char ch = 'x';  
int j = 77;  
double d = 6.02;  
string str1 = "How are u? ";  
string str2 = "Pretty Good";
```

11.6.1 code

11.7 String with Embedded blanks

Write a program that creates a file “**test.txt**” and write the following lines in that file

If we don't work hard,
We will not be able to score high &
might some our class fellow get flunked
repeating the semester again

Write these lines one by one in the same format given above

11.7.1 Code

Lab Evaluation Summary

Lab #	Lab Topics	Date conducted	Total Marks	Marks Obtained	Signature
1	Basic				
2	If, if-else, Nested if else				
3	Switch Statement				
4	For loop, While loop				
5	Do- while loop				
6	Function				
7	Arrays, Strings				
8	Two Dimensional Arrays				
9	Structures				
10	Pointers				
11	Files				
Total					

Note: It is the responsibility of every student to get this sheet signed from his/her lab engineer after every lab.